Analytical Dyanamics Final Project Report - Motion of a Top

1. Description of Motion of a Top

In rigid body dynamics, a symmetrical top is defined as a body with an axis of rotational symmetry that is subjected to a uniform gravitational field and supported at a single fixed point on its symmetrical axis. Because the body is constrained at one point (Fixed point O), its translational motion is restricted, leaving it with 3 degrees of freedom. These are described using Euler angles as defined as below and Figure 1.

1. **Precession** is the rotation of the top's axis around the fixed vertical axis.
2. **Nutation** is the tipping angle of the top's symmetry axis relative to the horizontal/vertical plane.
3. **Spin** is the top's rotation around its own axis of symmetry.

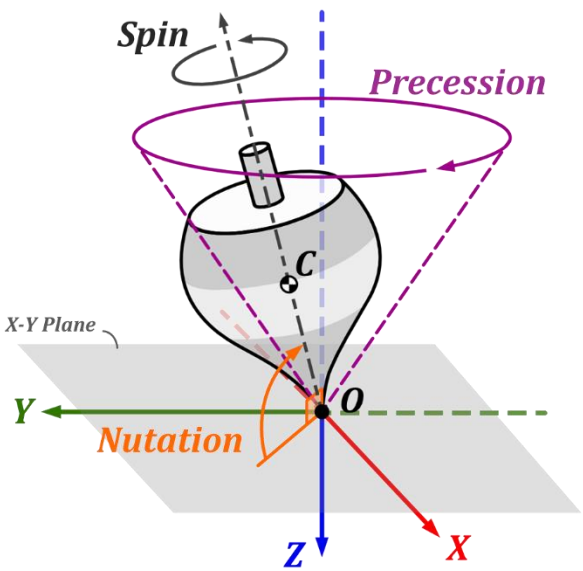


Figure 1 A symmetrical top and Euler angles

The resulting motion is a complex interplay between the gravitational torque trying to pull the top down and the gyroscopic forces generated by its spin, which cause it to sweep horizontally and oscillate vertically.

2. Derivation of Lagrange Equation of Motion

We define two reference frames are fixed frame (X, Y, Z) and a body-fixed frame (x, y, z) attached to the top as shown in Figure 2, where the z-axis represents the axis of rotational symmetry. The transformation between these frames is a sequence of Euler angles as defined in section 1.

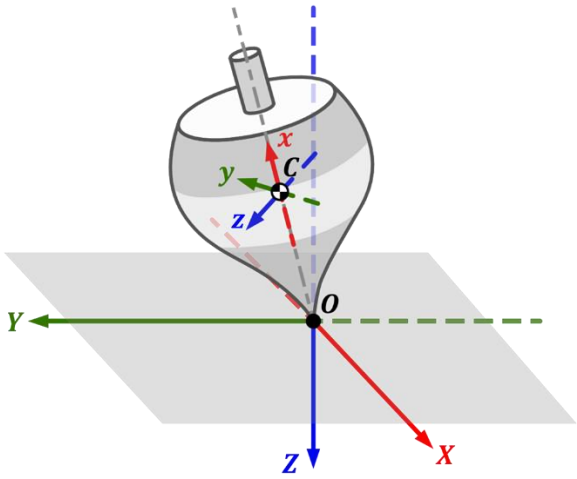


Figure 2 Reference frames of a symmetrical top

Sine this system has 3-DOF of motion is Euler angles. So, the generalized coordinate, q_i is a set of $\{\psi, \theta, \phi\}$

2.1 Kinetic Energy

The total kinetic energy of the top is only rotational because the apex is pinned at a fixed point O. The rotational kinetic energy is given by

$$T = T_{rot} = \frac{1}{2}\left(I_{xx}\omega_x^2 + I_{yy}\omega_y^2 + I_{zz}\omega_z^2\right) \tag{1}$$

let x-axis be symmetrical axis of the top. the moment of inertia about the symmetry axis is the axial inertia ($I_{xx} = I_a$) and the principal moments of inertia along the transverse axes are equal ($I_{yy} = I_{zz} = I_t$)

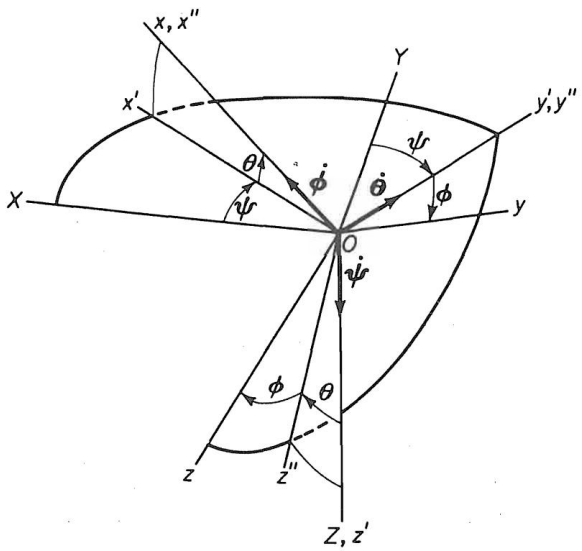


Figure 3 Euler angle components of the angular velocity of the xyz system

According to Figure 3, If we project the Euler angle rates $(\dot{\psi}, \dot{\theta}, \dot{\phi})$ onto the body-fixed principal axes and separate the angular velocity components. The transverse angular velocity squared $(\omega_y^2 + \omega_z^2)$ is consists of nutation rate and the projection of precession rate onto the transverse plane.

$$\omega_y = \dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi \tag{2}$$

$$\omega_z = \dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi \tag{3}$$

$$\omega_y^2 + \omega_z^2 = \dot{\theta}^2 + \dot{\psi}^2\cos^2\theta \tag{4}$$

The axial angular velocity (ω_x) is the sum of the relative spin and the projection of the precession rate onto the symmetry axis.

$$\boxed{\omega_x = \Omega = \dot{\phi} - \dot{\psi}\sin\theta} \tag{5}$$

Substituting (2) and (3) back to (1) we will obtain total kinetic energy

$$T = \frac{1}{2}\left(I_t\omega_x^2 + I_t\omega_y^2 + I_a\omega_z^2\right) = \frac{1}{2}\left(I_t\left(\omega_x^2 + \omega_y^2\right) + I_a\omega_z^2\right)$$

$$T = \frac{1}{2}\left(I_t\left(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta\right) + I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2\right)$$

$$\boxed{T = \frac{1}{2}I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + \frac{1}{2}I_t\left(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta\right)} \tag{6}$$

2.2 Potential Energy

The potential energy is determined by the elevation of the center of mass. Let l is the distance from the point O to the center of mass C, shown in Figure 2, then the vertical height is $h = l\sin\theta$. So, the potential energy is as shown

$$V = mgl\sin\theta \tag{7}$$

2.3 Lagrange's Equations

The Lagrangian equation is as shown

$$L = T - V$$

$$\boxed{L = \frac{1}{2}I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + \frac{1}{2}I_t\left(\dot{\theta}^2 + \dot{\psi}^2\cos^2\theta\right) - mgl\sin\theta} \tag{8}$$

To find the equations of motion, we apply the Euler-Lagrange to each generalized coordinate $q_i \in \{\psi, \theta, \phi\}$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \left(\frac{\partial L}{\partial q_i}\right) = 0 \tag{9}$$

From Equation (8) we will see that the spin (ϕ) and the precession (ψ) do not appear in the Lagrangian. Therefore, there are ignorable coordinates, meaning its conjugate momentum is conserved. So, we can write:

$$p_\psi = \frac{\partial L}{\partial \dot{\psi}} = -I_a \Omega \sin\theta + I_t \dot{\psi} \cos^2\theta \ = \text{constant} \tag{10}$$

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right) = I_a \Omega = \text{constant} \tag{11}$$

Rearrange Equation (10), we can express the dynamic precession rate entirely as a function of the nutation angle as shown

$$\boxed{\dot{\psi} = \frac{p_\psi + I_a \Omega \sin\theta}{I_t \cos^2\theta}} \tag{12}$$

Rearrange Equation (5), we can express the spin rate as shown

$$\boxed{\dot{\phi} = \Omega + \dot{\psi}\sin\theta} \tag{13}$$

However, the nutation (θ) appears in both the potential energy and the geometric projections of the kinetic energy. Applying the Euler-Lagrange in (9) to θ we will get the differential equation.

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = I_t \dot{\theta} \tag{14}$$

$$\frac{\partial L}{\partial \theta} = -I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta - I_t\dot{\psi}^2 \sin\theta\cos\theta - mgl\cos\theta \tag{15}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \left(\frac{\partial L}{\partial \theta}\right) = 0$$

$$I_t\ddot{\theta} + I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta + I_t\dot{\psi}^2\sin\theta\cos\theta + mgl\cos\theta = 0 \tag{16}$$

Rearrange equation (16) we will get non-linear second-order ODE for the top's vertical motion.

$$\boxed{\ddot{\theta} = \frac{-I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta - I_t\dot{\psi}^2\sin\theta\cos\theta - mgl\cos\theta}{I_t}} \tag{17}$$

3. Numerical Simulation Setup

To simulate the dynamics of the top computationally, the second-order Lagrangian differential equations should be reduced into a system of first-order ODE. We define a state vector ($\boldsymbol{S}$), consisting of 4 variables are 3 Euler angles and the rate of change of the nutation angle ($\dot{\theta}$). The precession rate ($\dot{\psi}$) and relative spin rate ($\dot{\phi}$) do not need to be stored as independent states because they can be algebraically reconstructed from the conserved momenta and the current nutation angle.

$$\boldsymbol{S} = [S_1, S_2, S_3, S_4]^T = \left[\theta, \psi, \phi, \dot{\theta}\right]^T \tag{18}$$

The First-Order Derivative Vector will be:

$$\dot{\boldsymbol{S}} = \left[\dot{S}_1, \dot{S}_2, \dot{S}_3, \dot{S}_4\right]^T \tag{19}$$

Where:

$$\dot{S}_1 = \dot{\theta} = S_4$$

$$\dot{S}_2 = \dot{\psi} = \frac{p_\psi + I_a \Omega \sin\theta}{I_t \cos^2\theta}$$

$$\dot{S}_3 = \dot{\phi} = \Omega + \dot{\psi}\sin\theta$$

$$\dot{S}_4 = \ddot{\theta} = \frac{-I_a\left(\dot{\phi} - \dot{\psi}\sin\theta\right)\dot{\psi}\cos\theta - I_t\dot{\psi}^2\sin\theta\cos\theta - mgl\cos\theta}{I_t}$$

When setting up this system for a numerical solver, we must handle the singularity in the $\dot{\psi}$ equation ($\dot{S}_2$). When the top is perfectly horizontal ($\theta = 90°$). At this angle, the math tries to divide by zero, which will instantly crash the simulation. So, we add a safety net to the code. If it gets dangerously close to zero ($\sim 10^{-6}$), this lets the computer step safely past the error without messing up the actual physics.

The physical parameters that we use in this simulation are shown in Table 1, and all of simulation cases will use same time span of 10 seconds.

Table 1 The physical parameters used in simulation

Physical Parameters	Symbol	Value	Units
Mass of the top	m	1.0	kg
Distance between point O to point C	l	1.0	m
Gravitational acceleration	g	9.81	$\mathrm{m/s^2}$
Axial Moment of Inertia	I_a	1.0	$\mathrm{kg.m^2}$
Transverse Moment of Inertia	I_t	5.0	$\mathrm{kg.m^2}$

4. Simulation Case

Instead of dealing with complicated angles, the textbook uses the letter u to represent the vertical height of the top's tip.

$$u = \sin\theta \tag{20}$$

Using angles like θ directly makes the energy equations messy. Swapping to u turns that complex trigonometry into a simpler math formula. u_1 and u_2 is physical boundaries (the lowest and highest points) and u_0 is the exact height where the top stops sweeping forward (precession rate equals zero).

Comparing where u_0 falls relative to u_1 and u_2 can predict if the top's path will make wavy lines, loops, or sharp points as shown in Figure 4.

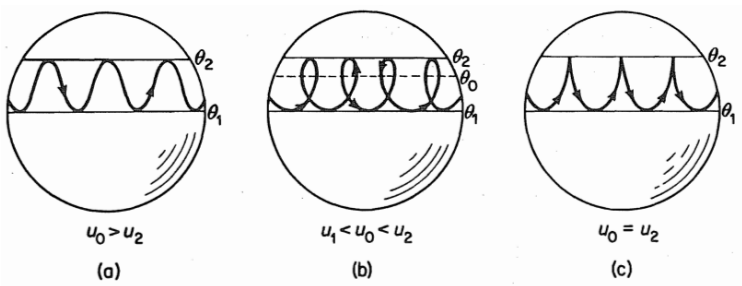


Figure 4 Possible paths of the axis of symmetry

4.1 Generating Paths as Shown in Figure 4 (Fig. 8-15 in Textbook)

In textbooks classify the generalized motion of the top based on the initial precession velocity and its relation to the critical turning parameter $\left(u_0 = -\frac{p_\psi}{p_\phi}\right)$ to three cases. We set up cases with initial condition as follows, where $\dot{\theta} = 0$ for all of cases.

1. **Case (a)** $u_0 > u_2$

   $\theta(0) = 60°, \quad \dot{\psi}(0) = -1.0 \text{ rad/s}, \quad \Omega = 12.0 \text{ rad/s}$

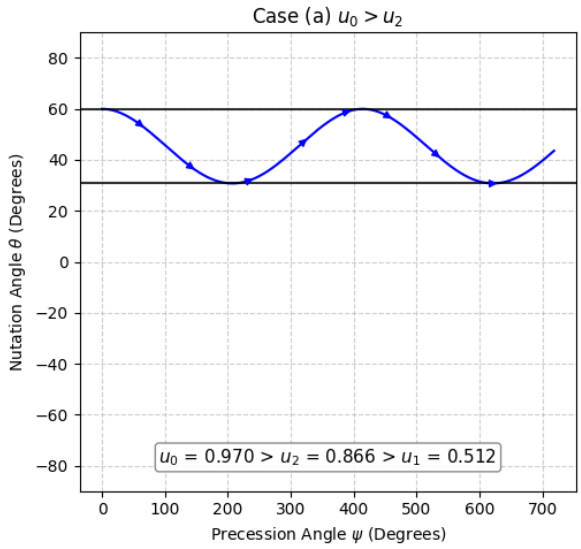
2. **Case (b)** $u_1 < u_0 < u_2$

   $\theta(0) = 60°, \quad \dot{\psi}(0) = 1.0 \text{ rad/s}, \quad \Omega = 15.0 \text{ rad/s}$

3. **Case (c)** $u_0 = u_2$

   $\theta(0) = 60°, \quad \dot{\psi}(0) = 0.0 \text{ rad/s}, \quad \Omega = 15.0 \text{ rad/s}$

The Result of Possible path of the axis of symmetry and 3D trajectory plots of cases (a), (b) and (c) is shown in Figure 5, 6 and 7 respectively.



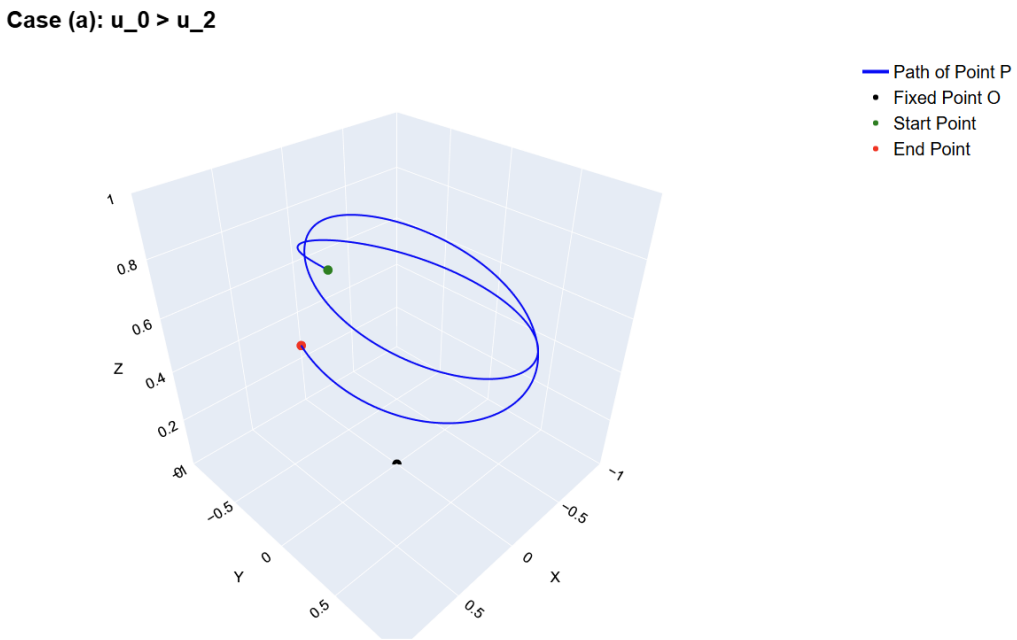
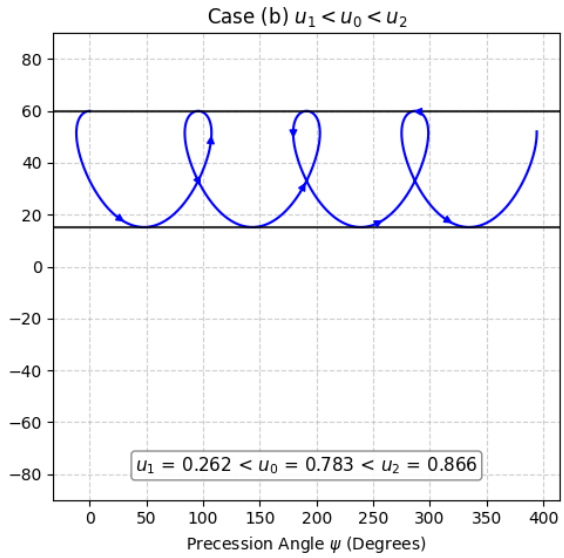


Figure 5 Possible path of the axis of symmetry and 3D trajectory plots of cases (a) $u_2 < u_0$

In this case, $u_0 = 0.970$ is higher than the highest point the top can reach $u_2 = 0.866$. Because the top never rises high enough to hit this stopping line, its forward sweeping speed never drops to zero. As a result, the top just keeps moving forward in a smooth, continuous wave, bouncing gently between 60 degrees and about 31 degrees.



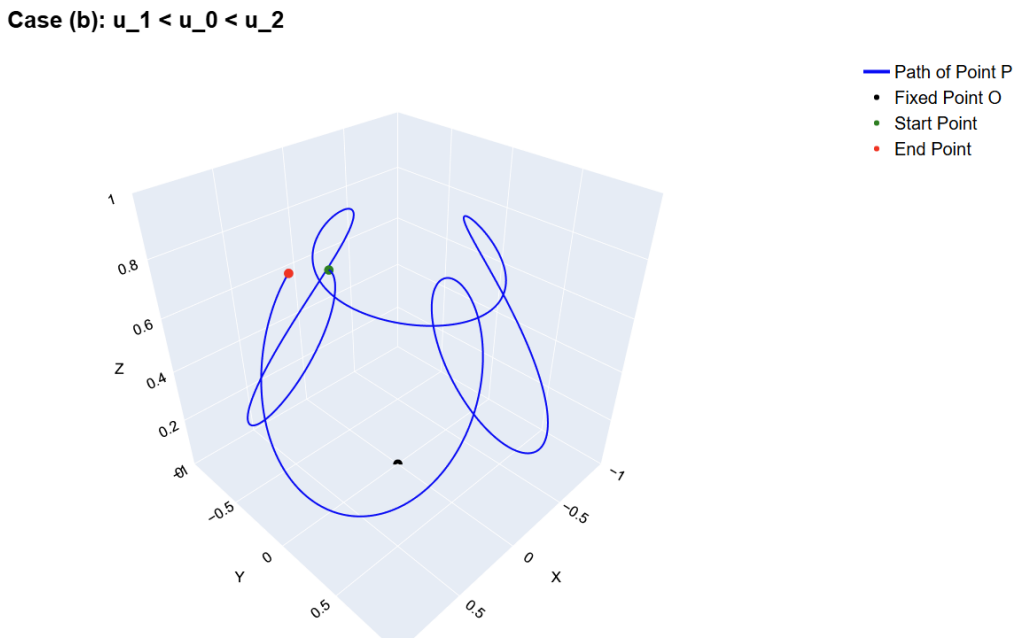
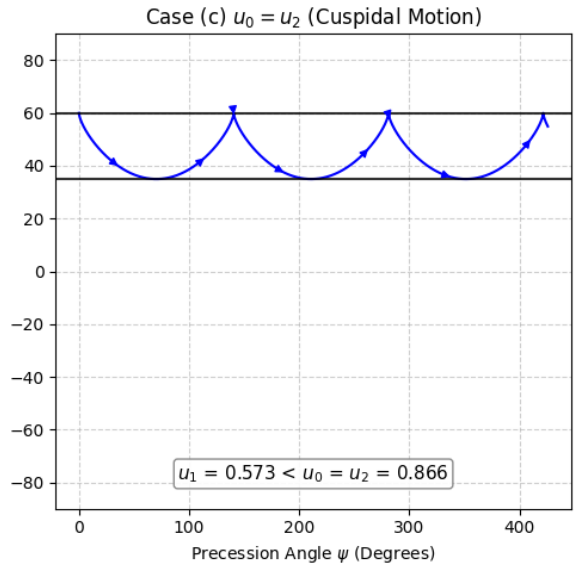


Figure 6 Possible path of the axis of symmetry and 3D trajectory plots of cases (b) $u_1 < u_0 < u_2$

In this case, $u_0 = 0.783$ is in the middle of the top's upper and lower bouncing limits. This means that as the top nods up and down, it crosses this zero-speed line. When it crosses the line, the top actually reverses direction for a split second. This brief backward sweep is what creates the distinct curly loops at the top of the graph before the top falls down to about 15 degrees.



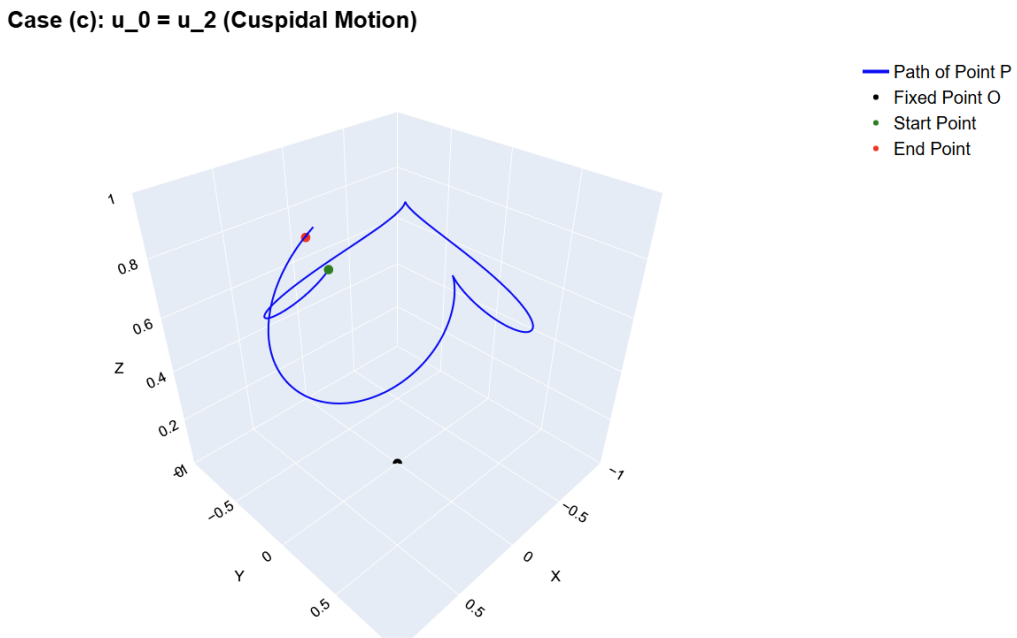


Figure 7 Possible path of the axis of symmetry and 3D trajectory plots of cases (c) $u_0 = u_2$

In this case, the top is dropped from a dead standstill, u_0 is exactly equal to its starting height ($u_2 = 0.866$). Every time the top bounces back up to its peak at 60 degrees, it hits this exact stopping point and remains still for a fraction of a second before falling again. This sudden stop-and-drop creates the sharp corners (cusps) seen on the graph.

4.2 Both Case of Cuspidal Motion (Pages 426-427 in Textbook)

We simulate two specific boundary conditions for cuspidal motion as follows:

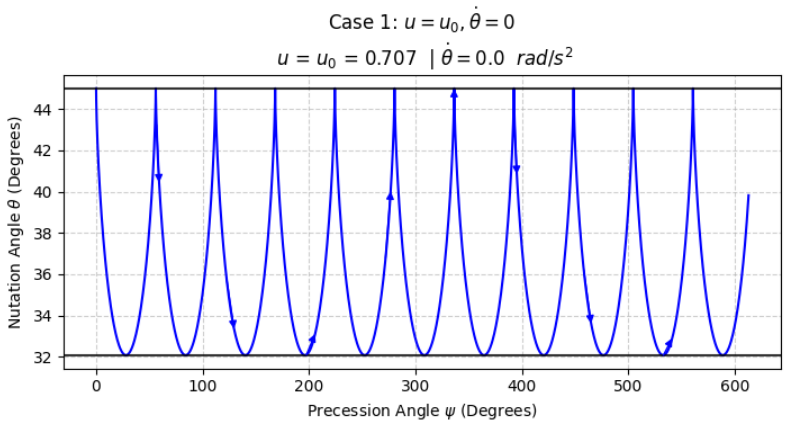
1. **Case 1:** $u = u_0$ when $\dot{\theta} = 0$

   $\theta(0) = 45°, \quad \dot{\theta}(0) = 0 \text{ rad/s}, \quad \dot{\psi}(0) = 0 \text{ rad/s}, \quad \Omega = 20.0 \text{ rad/s}$

2. **Case 2:** $u = u_0 = 1$ when $\dot{\theta} \neq 0$

   $\theta(0) = 90°, \quad \dot{\theta}(0) = 1.0 \text{ rad/s}, \quad \dot{\psi}(0) = 0 \text{ rad/s}, \quad \Omega = 20.0 \text{ rad/s}$

The Result of Possible path of the axis of symmetry and 3D trajectory plots of case 1 and 2 is shown in Figure 8 and 9 respectively.



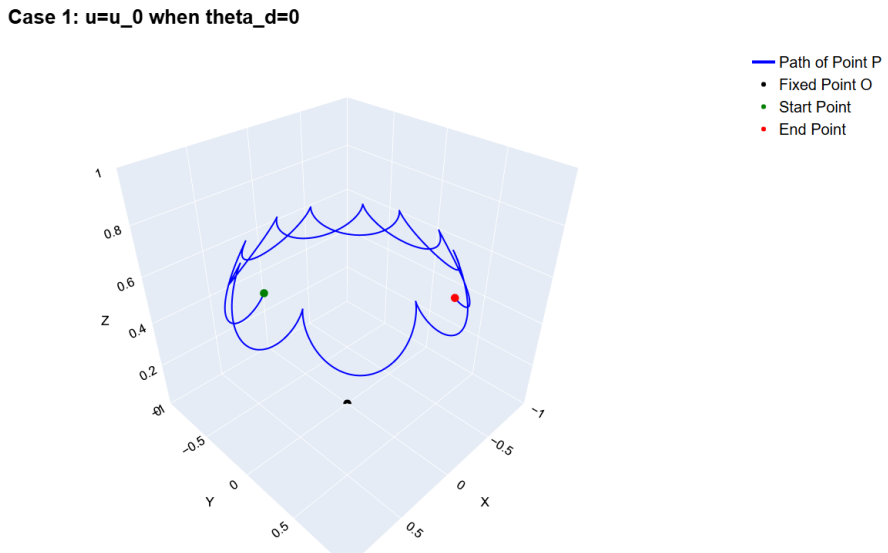


Figure 8 Possible path of the axis of symmetry and 3D trajectory plots of case 1: $u = u_0$ when $\dot{\theta} = 0$

In this case, the top is dropped from a dead stop at a 45° ($u_0 = 0.707$). Because it is released at zero speed, every time it bounces back up to its starting height of 45°, it runs completely out of upward momentum. It freezes for a split second before falling back down. This complete stop-and-drop cycle is exactly what creates the sharp, pointy corners (cusps) along the top boundary of the graph.

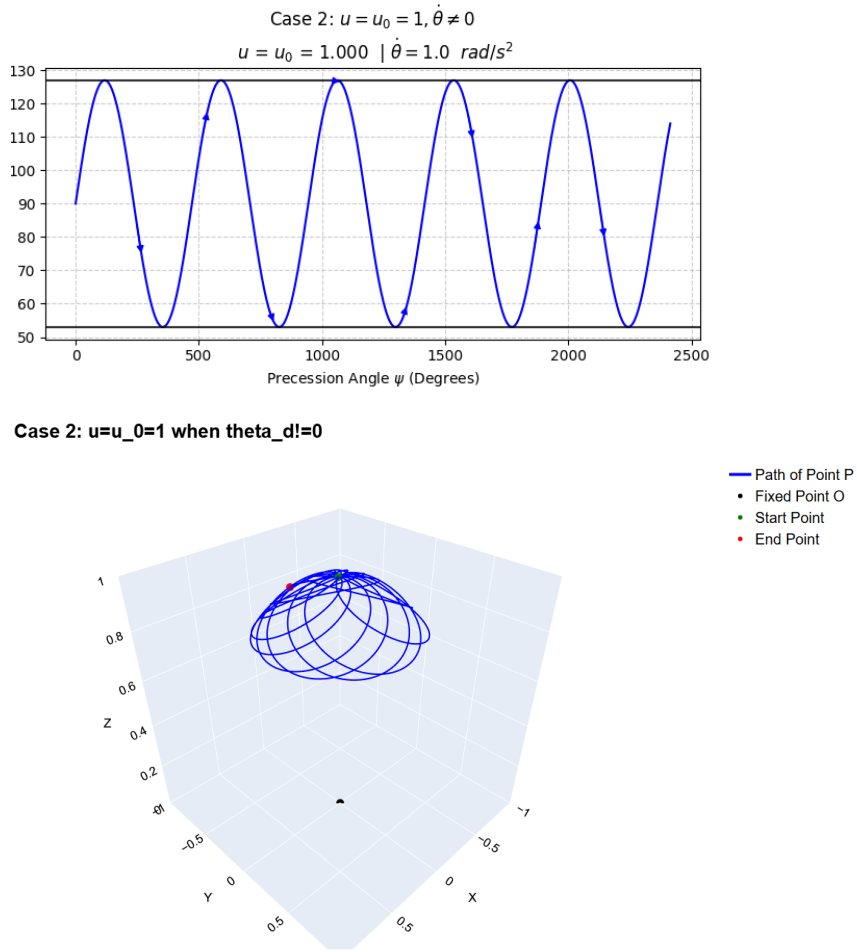


Figure 9 Possible path of the axis of symmetry and 3D trajectory plots of case 2: $u = u_0 = 1$ when $\dot{\theta} \neq 0$

In this case, the top starts at the vertical axis. but instead of being dropped from rest, it is given an initial downward push. Because it is already carrying speed when it crosses this boundary, it never stops forming a sharp point. Its momentum carries it straight through the horizontal plane. As shown in the graph, the top smoothly waves up and down swinging deeply between about 53° and 127° without ever freezing in place or forming a cusp.

4.3 Stability of motion near the vertical (Pages 428-429 in Textbook)

We simulate the top motion if the axis points vertically upward when $\dot{\theta} = 0$. In textbooks define stability parameter (λ) as shown in (21) and separate it into 3 cases. We setup conditions for each case as follows:

$$\lambda = \frac{I_a^2 \Omega^2}{4 I_t m g l} \tag{21}$$

Let $\Omega_{\text{critical}} = \sqrt{4 I_t m g l / I_a^2}$ is critical spin rate

1. **Case 1:** $\lambda < 1$ or $\Omega^2 < 4 I_t m g l / I_a^2$ or $\Omega < \Omega_{\text{critical}}$

   $\theta(0) = 85°$, $\dot{\theta}(0) = 0$ rad/s, $\dot{\psi}(0) = 0$ rad/s, $\Omega = 0.8\,\Omega_{\text{critical}}$

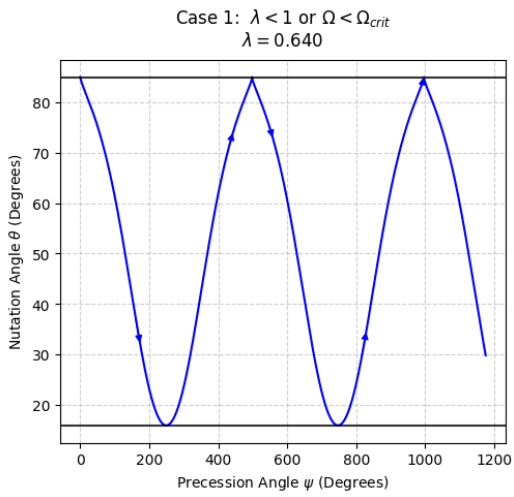
2. **Case 2:** $\lambda = 1$ or $\Omega^2 = 4 I_t m g l / I_a^2$ or $\Omega = \Omega_{\text{critical}}$

   $\theta(0) = 85°$, $\dot{\theta}(0) = 0$ rad/s, $\dot{\psi}(0) = 0$ rad/s, $\Omega = \Omega_{\text{critical}}$

3. **Case 3:** $\lambda > 1$ or $\Omega^2 > 4 I_t m g l / I_a^2$ or $\Omega > \Omega_{\text{critical}}$

   $\theta(0) = 85°$, $\dot{\theta}(0) = 0$ rad/s, $\dot{\psi}(0) = 0$ rad/s, $\Omega = 1.2\,\Omega_{\text{critical}}$

The Result of Possible path of the axis of symmetry and 3D trajectory plots of case 1, 2 and 3 is shown in Figure 10, 11 and 12 respectively.



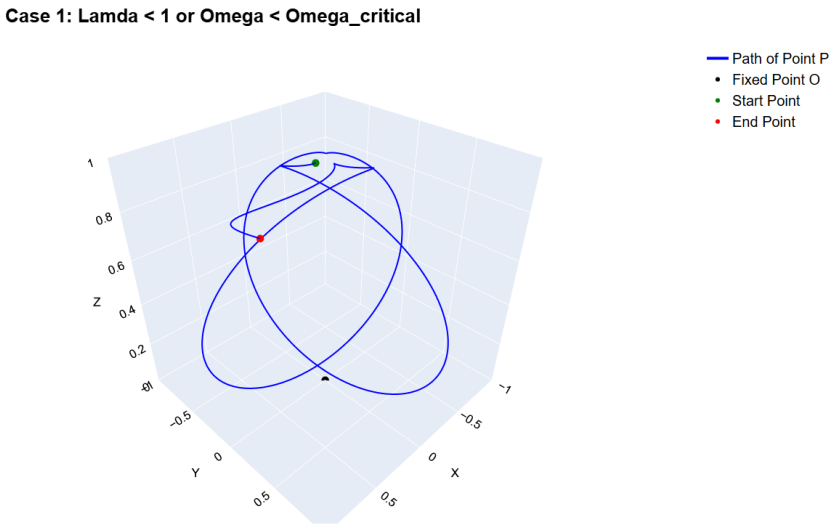


Figure 10 Possible path of the axis of symmetry and 3D trajectory plots of case 1: $\lambda < 1$ or $\Omega^2 < 4I_t mgl/I_a^2$ or $\Omega < \Omega_{\text{critical}}$

In this case, slow spin, the top does not have enough spin to fight gravity effectively. When released from 85 degrees, it suffers a massive fall, plummeting all the way down to about 16 degrees before its weak spin can finally catch it and push it back up.

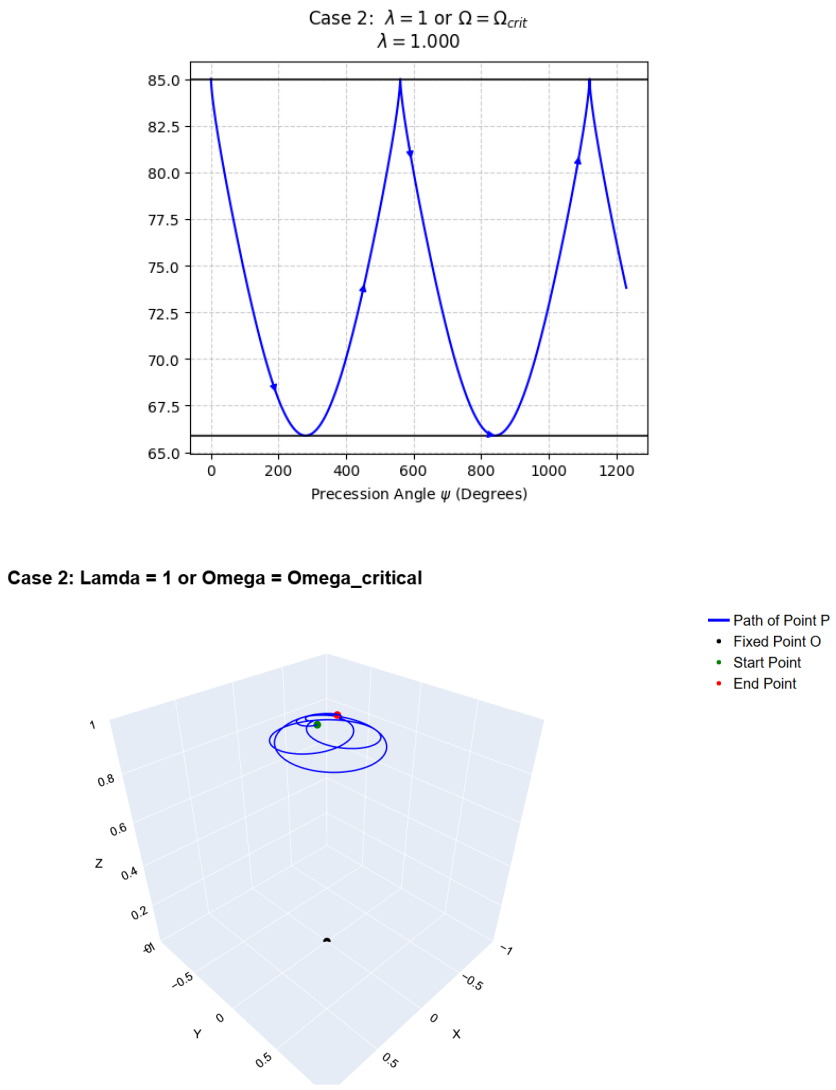


Figure 11 Possible path of the axis of symmetry and 3D trajectory plots of case 2: $\lambda = 1$ or $\Omega^2 = 4I_t mgl/I_a^2$ or $\Omega = \Omega_{\text{critical}}$

In this case, the top is spinning at the exact mathematical threshold where the gyroscopic forces and gravity start to balance. It still falls when released, but the spin catches it much sooner, limiting its downward dip to about 66°.

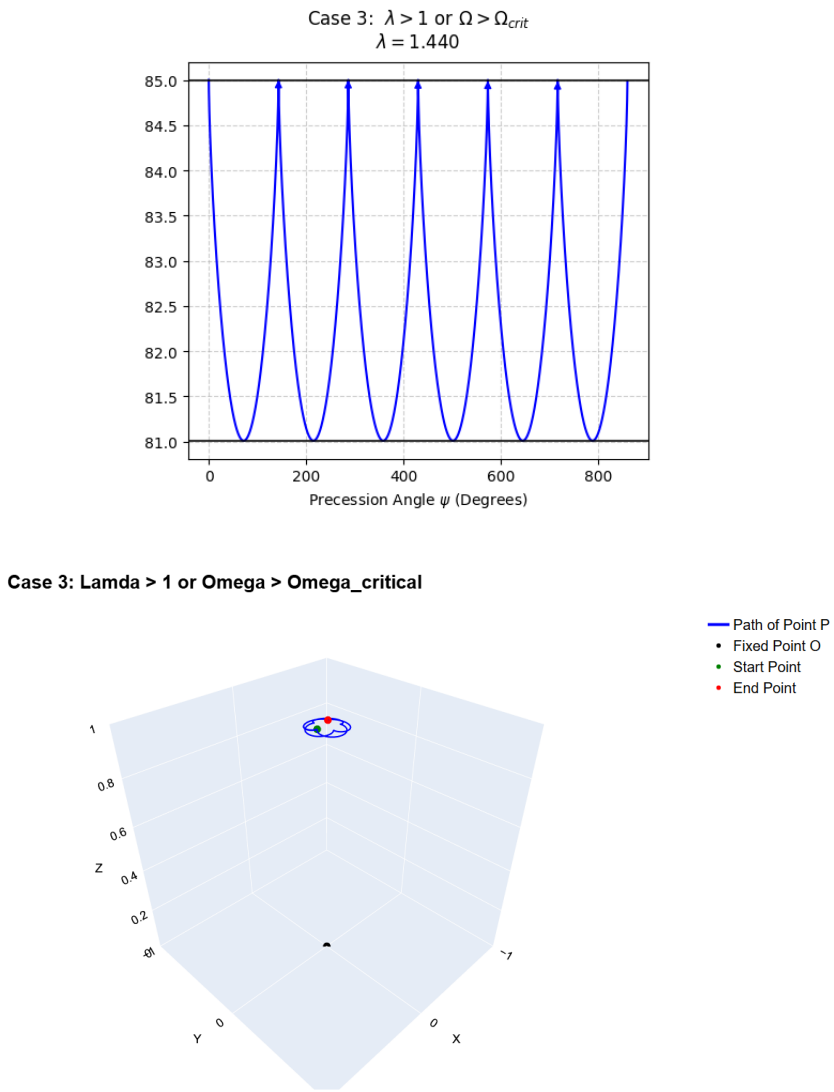


Figure 12 Possible path of the axis of symmetry and 3D trajectory plots of case 3: $\lambda > 1$ or $\Omega^2 > 4I_t mgl/I_a^2$ or $\Omega > \Omega_{\text{critical}}$

In this case, the top's spin is completely dominating gravity. When it dropped, it barely falls at all. The strong gyroscopic forces tightly lock its position, allowing it to only dip slightly down to 81 degrees before bouncing right back up in a very tight, controlled wobble.

4.4 Nutation and Amplitude for the case of large spin (Pages 429-430 in Textbook)

In textbook approximate amplitude and frequency of the nutation motion of a fast top ($\lambda \gg 1$) from roots of cuspidal motion case. The amplitude of the nutation as shown in (22) and circular frequency of the nutation as shown in (23)

$$u_0 - u_1 \cong \frac{1}{2\lambda}(1 - u_0^2) = \frac{2I_t mgl}{I_a^2 \Omega^2}(1 - u_0^2) \tag{22}$$

$$2p \cong \frac{I_a \Omega}{I_t} \tag{23}$$

We simulate motion of a fast top by separating it into 3 cases of different Ω values. We set conditions for each case as shown below:

1. **Case 1:** Moderate Spin

   $\theta(0) = 60°, \quad \dot{\theta}(0) = 0 \text{ rad/s}, \quad \dot{\psi}(0) = 0 \text{ rad/s}, \quad \Omega = 25 \text{ rad/s}$

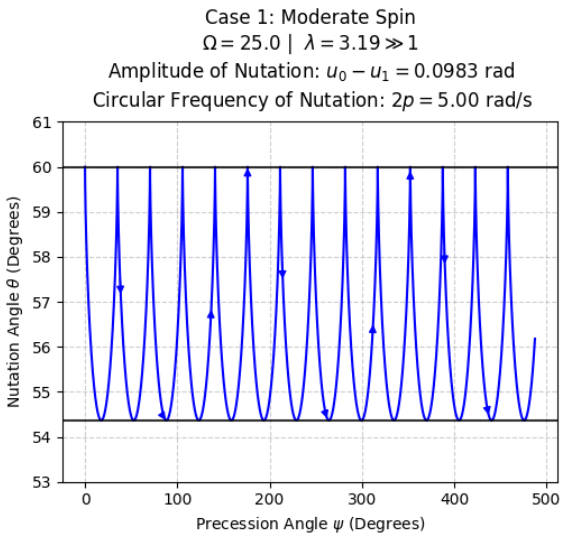
2. **Case 2:** Large Spin

   $\theta(0) = 60°, \quad \dot{\theta}(0) = 0 \text{ rad/s}, \quad \dot{\psi}(0) = 0 \text{ rad/s}, \quad \Omega = 50 \text{ rad/s}$

3. **Case 3:** Very Large Spin

   $\theta(0) = 60°, \quad \dot{\theta}(0) = 0 \text{ rad/s}, \quad \dot{\psi}(0) = 0 \text{ rad/s}, \quad \Omega = 100 \text{ rad/s}$

The Result of Possible path of the axis of symmetry and 3D trajectory plots of case 1, 2 and 3 is shown in Figure 13, 14 and 15 respectively.



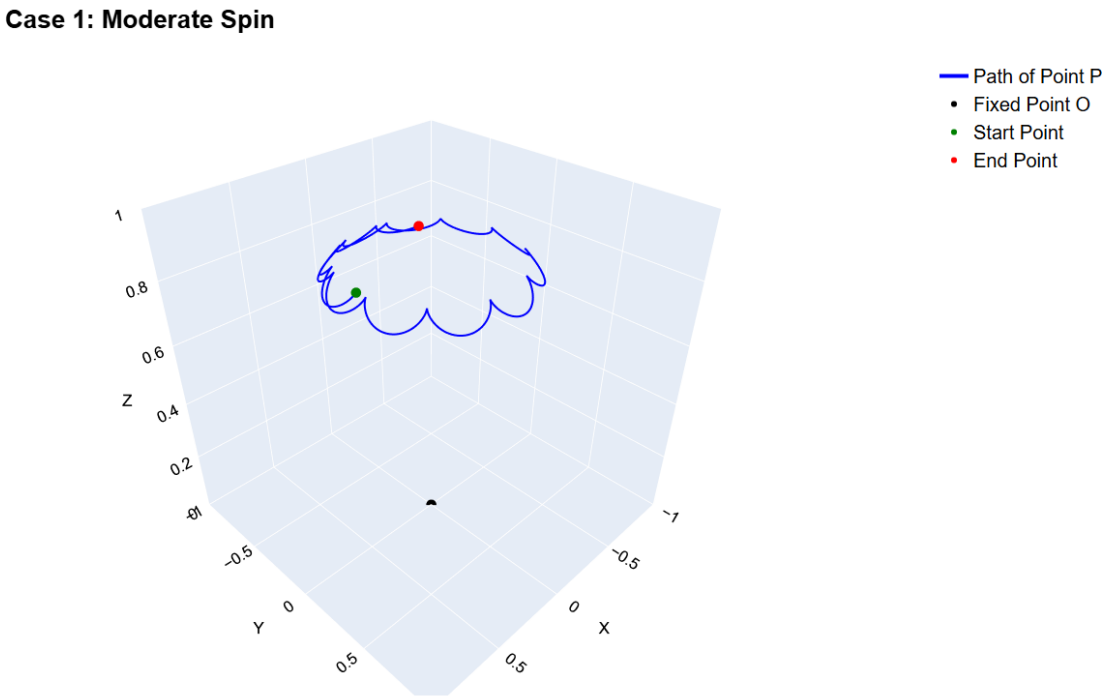
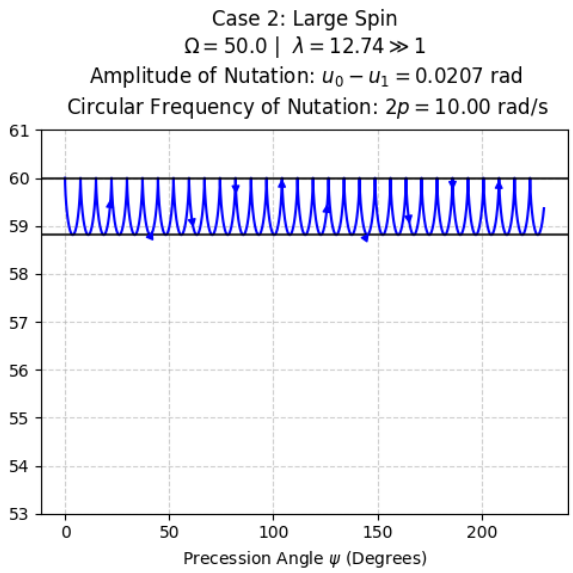


Figure 13 Possible path of the axis of symmetry and 3D trajectory plots of case 1: Moderate Spin

In this case, the top's bouncy motion is still clearly visible. It dips down a fair amount (amplitude of ~0.098 radians) with each bounce and completes those bounces at a relatively relaxed pace (frequency of 5 rad/s).



Case 2: Large Spin

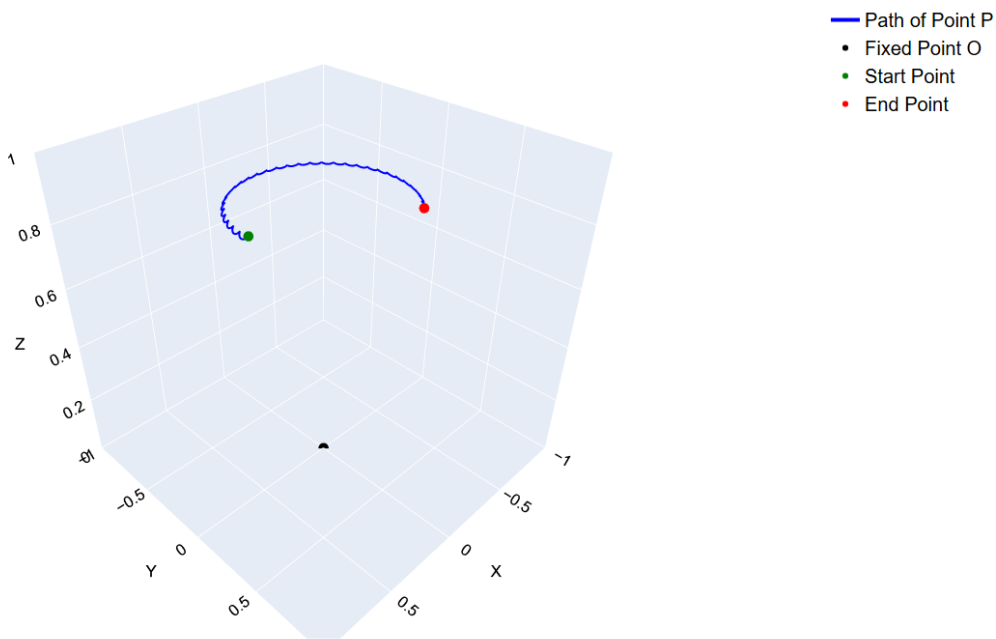
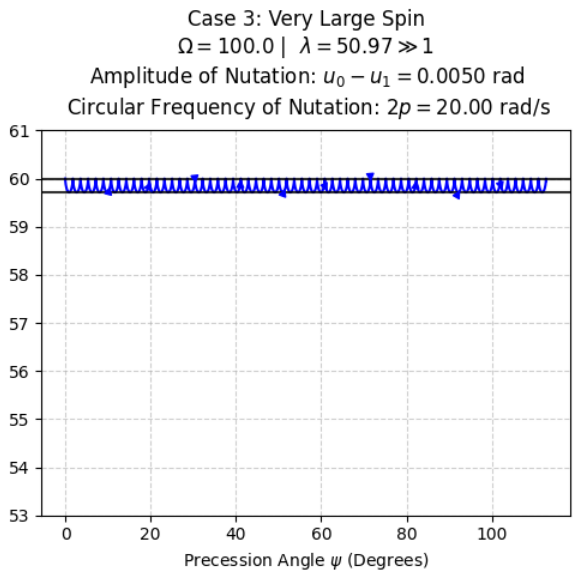


Figure 14 Possible path of the axis of symmetry and 3D trajectory plots of case 2: Large Spin

In this case, when we double the spin speed, the physics dramatically squash the wobble. The height of the bounces shrinks significantly (dropping to just ~0.02 radians), and the top vibrates twice as fast (10 rad/s) to keep up with the gyroscopic forces.



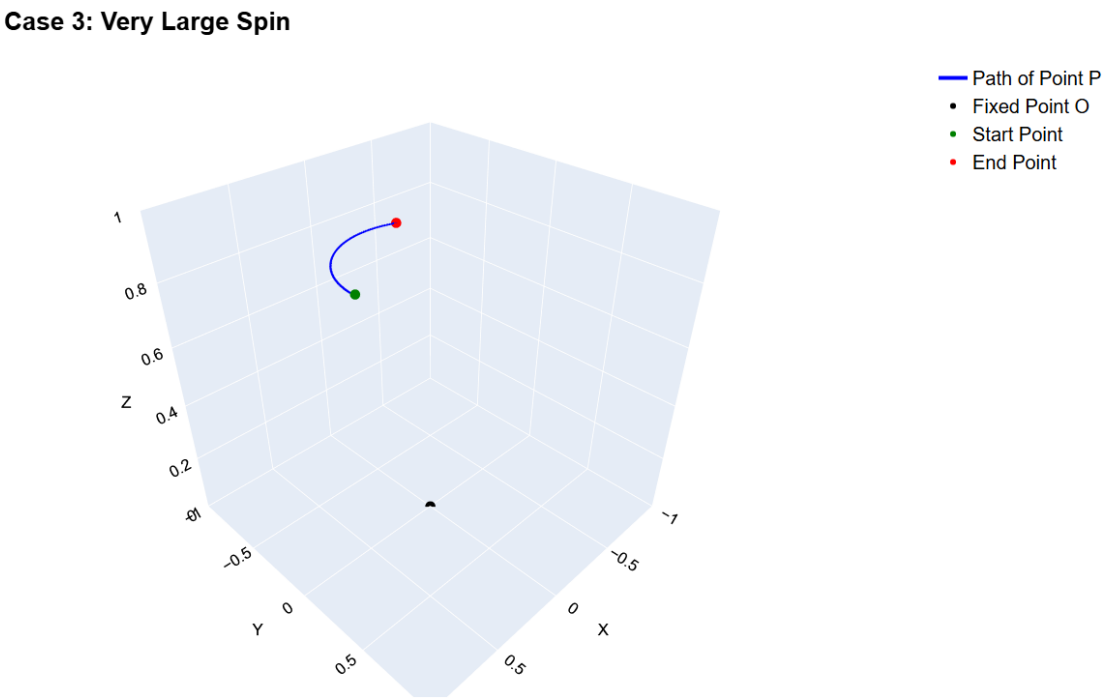


Figure 15 Possible path of the axis of symmetry and 3D trajectory plots of case 3: Very Large Spin

In this case, when we double the spin again practically crushes the wobble out of existence. The bounce amplitude is now microscopic (only ~0.005 radians), and it vibrates at a rapid 20 rad/s. Because these bounces are so tiny and happen so fast, the path appears as a flawlessly straight line in the real world. This effect is mathematically known as pseudo-regular precession.

4.5 Precession with no Nutation (Pages 431-432 in Textbook)

Precession with no nutation is indicated by having $u_1 = u_2$, that is, by the presence of a double root. In textbook they found the precession rate (In terms of θ) is as shown in (24) and they consider the case in which $\Omega^2 \gg 4I_t mgl/I_a^2$ that is, the spin is large enough so that the second term in the square root of (24) is small compared to unity and can approximate (24) by (25), where $b = I_a\Omega/I_t$ and $c = mgl/I_t$

$$\dot{\psi} = \frac{-b}{2\sin\theta}\left[1 \pm \sqrt{1 - \frac{2c\sin\theta}{b^2}}\right] \tag{24}$$

$$\dot{\psi} = \frac{-b}{2\sin\theta}\left[1 \pm \left(1 - \frac{c\sin\theta}{b^2}\right)\right] \tag{25}$$

We simulate precession with no nutation motion of the top using conditions are

$$\theta(0) = 60°, \qquad \dot{\theta}(0) = 0 \text{ rad/s}, \quad \Omega = 20 \text{ rad/s}$$

Then we can calculate $\dot{\psi}$ from (25), which will give us 2 values that are slow and fast roots. We use these values to run simulation and The Result of Possible path of the axis of symmetry of slow and fast roots is shown in Figure 15 and 3D trajectory plots of slow root and fast root are shown in Figure 16 and 17 respectively.

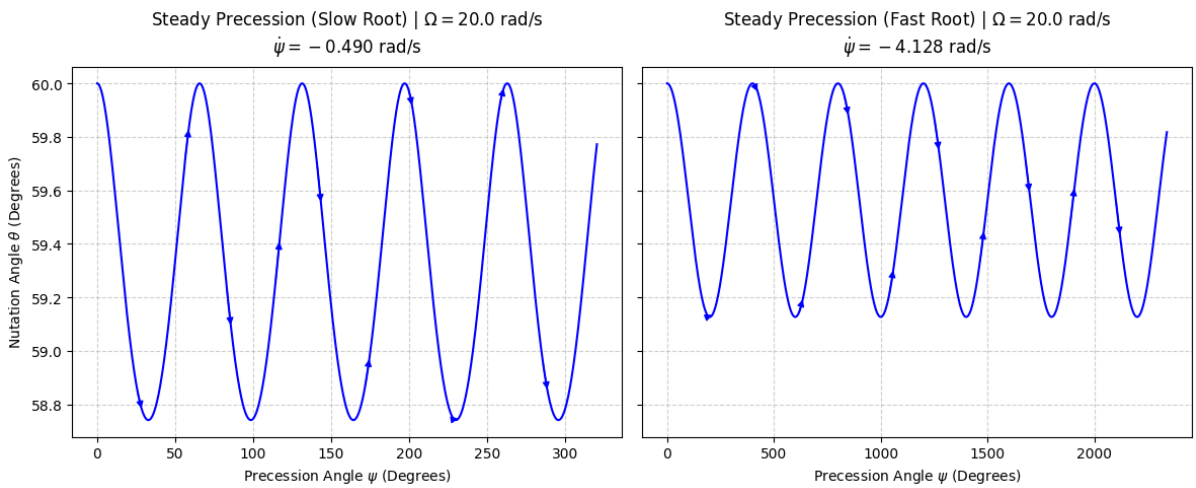


Figure 16 Possible path of the axis of symmetry of slow root (left) and fast root (right)

The simulation calculates a fast (-4.128 rad/s) and slow (-0.490 rad/s) sweeping speed intended to keep the top at a perfectly steady 60° height. However, the graphs reveal a tiny 1-degree wobble instead of a flat line. This happens because the simplified

formulas used in the code to calculate these starting speeds are slightly off from the exact mathematical ideal, causing the top to gently bounce as it sweeps.

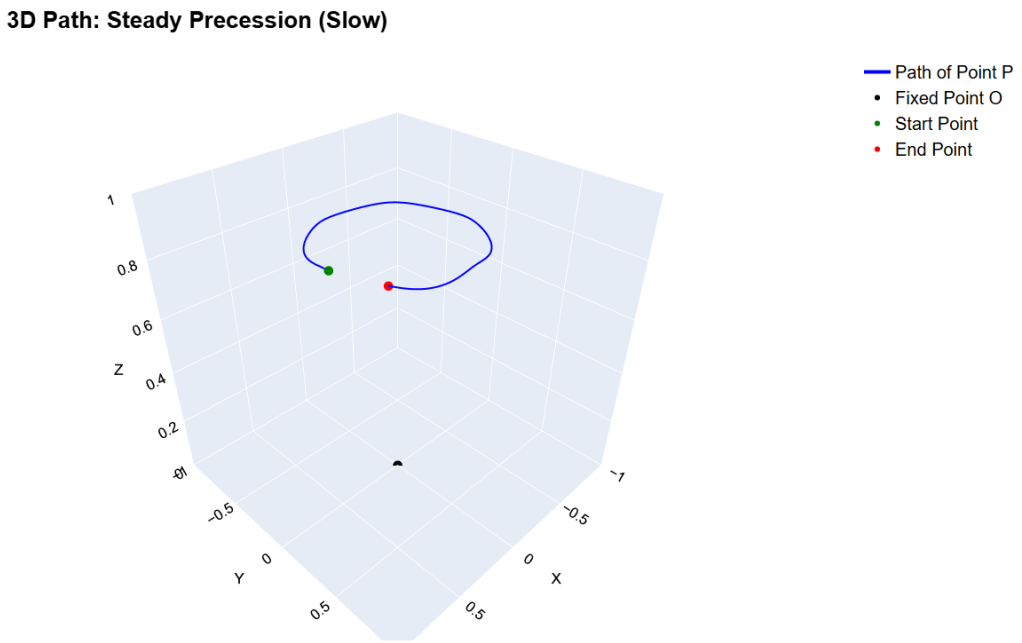


Figure 17 3D trajectory plots of slow root

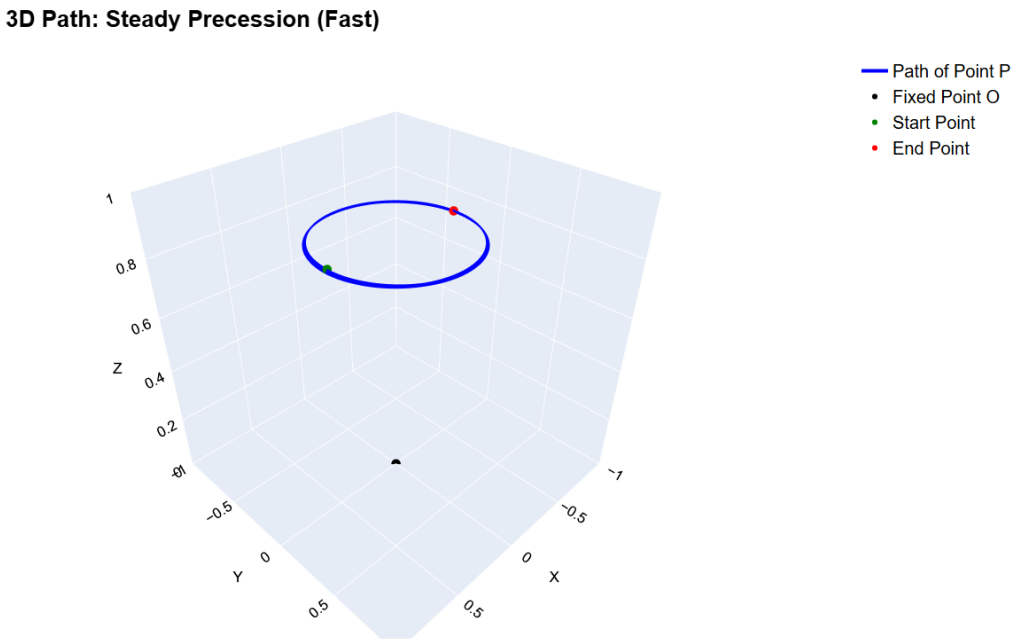


Figure 18 3D trajectory plots of fast root